DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 1

Data analyst

Profession 1 of 10 in Development and IT.

What the profession means

A data analyst answers business questions with data: pulls the records, cleans them, analyses patterns and communicates findings to decision makers.

What professionals do every day

Writes queries, builds reports and dashboards, checks data quality, runs comparisons and presents findings with a recommendation.

Why businesses hire them, and what problems they solve

Decisions need evidence. An analyst turns records into answers that managers can act on.

- Reports that take days to build by hand.
- Numbers that disagree between teams.
- Decisions made on averages that hide the detail.

Skills you need

- SQL and spreadsheets.
- Cleaning and validation.
- Charts that answer one question.
- Clear written findings.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Microsoft Excel	Analysis and reports	Free online version	Microsoft 365 plans	www.microsoft.com/microsoft-365/excel
Power BI Desktop	Dashboards	Free	Power BI Pro and Premium plans (verify)	powerbi.microsoft.com/desktop/
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Excel: basic formulas and a pivot table (Dt-B).
Intermediate	SQL: SELECT, GROUP BY, JOIN (Dt-C).

Advanced	Power BI: one dashboard (Dt-D).

Projects by level

- **Beginner:** Sales dashboard for a shop.
- **Intermediate:** Campaign comparison for a clinic.
- **Advanced:** Monthly KPI report for a firm.

Client-ready portfolio project

The sales dashboard project (Module G), with findings and a recommendation.

Worked example

Worked example. A query shows search earns the most per order; the analyst checks whether the number of orders or the order size explains it before recommending more budget.

Assignment with full solution

ASSIGNMENT

Freelance project: sales dashboard for a fictional shop

Business problem: A shop owner wants to see revenue by channel and by month, updated every week, without learning SQL.

Steps

1. Create the tables and load the sample data (see the SQL module).
2. Write the two queries: revenue by channel and monthly revenue.
3. Build a dashboard in Excel or Power BI with one chart per query.
4. Check each total against the hand calculation.
5. Write one paragraph of findings for the owner.

Expected output: A two-chart dashboard whose totals match the queries, and a findings paragraph.

SOLUTION (ANSWER KEY)

Channel totals: search 5,600; social 3,900; email 1,600 (sum 11,100). Monthly: January 3,700; February 2,100; March 5,300. Findings: search earns the most; February is the weakest month.

SQL

```
SELECT channel, SUM(amount) AS revenue
FROM orders
GROUP BY channel
ORDER BY revenue DESC;
```

SQL

```
SELECT substr(order_date, 1, 7) AS month, SUM(amount) AS revenue
FROM orders
```

```
GROUP BY month
ORDER BY month;
```

Common mistakes

- Reporting averages without the spread.
- Charts with truncated axes.
- Using numbers without checking the source.
- Recommendations with no evidence.

Freelancing opportunities

- Dashboards and monthly reports for small firms.
- Ad-hoc analysis requests on freelance marketplaces.
- Briefs for 'data analyst', 'Excel dashboard', 'SQL report'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Microsoft Learn: data analysis path
- SQLBolt (free interactive SQL lessons)

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.